

Math 241, F12
Exam 3

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally **3** problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	40	
2	30	
3	30	
Total	100	

Good Luck !

Problem 1. (40 points) Evaluate the following iterated integrals.

$$(a) \int_0^{\pi/4} \int_0^{\cos \theta} r \, dr d\theta.$$

$$\begin{aligned} &= \int_0^{\pi/4} \left[\frac{r^2}{2} \right]_{r=0}^{r=\cos \theta} d\theta \\ &= \int_0^{\pi/4} \frac{\cos^2 \theta}{2} d\theta = \int_0^{\pi/4} \frac{1}{2} \left[\frac{1 + \cos 2\theta}{2} \right] d\theta \\ &= \frac{1}{4} \left[\theta + \frac{\sin 2\theta}{2} \right]_{\theta=0}^{\theta=\pi/4} \\ &= \frac{1}{4} \cdot \left[\frac{\pi}{4} + \frac{\sin(\pi/2)}{2} - 0 \right] = \frac{\pi}{16} + \frac{1}{8} \end{aligned}$$

$$(b) \int_0^1 \int_0^y \int_0^{x+y} 2xy \, dz dx dy$$

$$\begin{aligned} &= \int_0^1 \int_0^y 2xy(x+y) \, dx dy \\ &= \int_0^1 \int_0^y (2x^2y + 2xy^2) \, dx dy \\ &= \int_0^1 \left[\frac{2}{3} x^3 y + x^2 y^2 \right]_{x=0}^{x=y} dy \\ &= \int_0^1 \left(\frac{2}{3} y^4 + y^4 \right) dy = \frac{5}{3} \int_0^1 y^4 dy \\ &= \frac{5}{3} \left[\frac{y^5}{5} \right]_{y=0}^{y=1} = \frac{1}{3} \end{aligned}$$

$$(c) \int_0^1 \int_0^z \int_0^y z e^{y^2} \, dx dy dz$$

$$\begin{aligned} &= \int_0^1 \int_0^z z y e^{y^2} \, dy dz \\ &= \int_0^1 \left[z \frac{e^{y^2}}{2} \right]_{y=0}^{y=z} dz \\ &= \int_0^1 z \left[\frac{e^{z^2}}{2} - \frac{1}{2} \right] dz \\ &= \int_0^1 \frac{z}{2} e^{z^2} - \frac{1}{2} z \, dz = \left[\frac{e^{z^2}}{4} - \frac{z^2}{4} \right]_{z=0}^{z=1} \\ &= \left[\frac{e}{4} - \frac{1}{4} \right] - \left[\frac{1}{4} - 0 \right] = \frac{e}{4} - \frac{1}{2} \end{aligned}$$

Problem 2. (30 points)

(a) Using the **cylindrical coordinates** to compute the triple integral $\iiint_E \sqrt{x^2 + y^2} dV$ where E is the solid that lies inside the cylinder $x^2 + y^2 = 4$, and between the plane $z = 0$ and the surface $z = 2 + x^2 + y^2$.

Solution: $x^2 + y^2 = r^2 \rightarrow \sqrt{x^2 + y^2} = r$

$$\begin{aligned} E &= \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, 0 \leq r \leq 2, 0 \leq z \leq 2 + r^2\} \\ \Rightarrow \iiint_E \sqrt{x^2 + y^2} dV &= \int_0^{2\pi} \int_0^2 \int_0^{2+r^2} r \cdot r dz dr d\theta \\ &= \int_0^{2\pi} \int_0^2 r^2 (2 + r^2) dr d\theta \\ &= \int_0^{2\pi} \left[\frac{2}{3} r^3 + \frac{r^5}{5} \right]_{r=0}^{r=2} d\theta \\ &= \int_0^{2\pi} \frac{16}{3} + \frac{32}{5} d\theta \\ &= \frac{176}{15} \cdot 2\pi = \frac{352}{15} \pi \end{aligned}$$

(b) Let H be the hemisphere defined by $x^2 + y^2 + z^2 \leq 9$, $z \geq 0$ with a density function $f(x, y, z) = z$. Find the mass of the solid H by evaluating $\iiint_H z dV$ using the triple integral in the **spherical coordinates**.

Solution: $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$, $z = \rho \cos \phi$.

$$\begin{aligned} H &= \{(\rho, \theta, \phi) \mid 0 \leq \rho \leq 3, 0 \leq \theta \leq 2\pi, 0 \leq \phi \leq \frac{\pi}{2}\} \\ x^2 + y^2 + z^2 &= \rho^2 \\ \Rightarrow \iiint_H z dV &= \int_0^3 \int_0^{2\pi} \int_0^{\pi/2} \rho \cos \phi \cdot \rho^2 \sin \phi d\phi d\theta d\rho \\ &= \left(\int_0^3 \rho^3 d\rho \right) \left(\int_0^{\pi/2} \cos \phi \sin \phi d\phi \right) \left(\int_0^{2\pi} d\theta \right) \\ &= \left(\left[\frac{\rho^4}{4} \right]_{\rho=0}^{\rho=3} \right) \left(\left[\frac{\sin^2 \phi}{2} \right]_{\phi=0}^{\phi=\pi/2} \right) (2\pi) \\ &= \frac{81}{4} \cdot \frac{1}{2} \cdot 2\pi \\ &= \frac{81}{4} \pi \end{aligned}$$

Problem 3. (30 points) Let C be the space curve defined by the vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$ with

$$x = t, \quad y = 3 \cos t, \quad z = 3 \sin t, \quad 0 \leq t \leq \pi.$$

(a) Evaluate the line integral $\int_C yz \cos(x) \, ds$.

$$\begin{aligned} \text{Solution: } \int_C yz \cos(x) \, ds &= \int_0^\pi (3 \cos t)(3 \sin t)(\cos t) \sqrt{1 + (-3 \sin t)^2 + (3 \cos t)^2} \, dt \\ &= \int_0^\pi 9 \cos^2 t \sin t \sqrt{1 + 9 \sin^2 t + 9 \cos^2 t} \, dt \\ &= 9\sqrt{10} \int_0^\pi \cos^2 t \sin t \, dt \\ &= 9\sqrt{10} \left[-\frac{\cos^3 t}{3} \right]_{t=0}^{t=\pi} \\ &= 9\sqrt{10} \cdot \left[\frac{1}{3} + \frac{1}{3} \right] \\ &= 6\sqrt{10} \end{aligned}$$

(b) Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y, z) = \langle z, y, -x \rangle$.

$$\begin{aligned} \text{Solution: } \int_C \mathbf{F} \cdot d\mathbf{r} &= \int_0^\pi \mathbf{F}(t) \cdot \mathbf{r}'(t) \, dt \\ &= \int_0^\pi \langle 3 \sin t, 3 \cos t, -t \rangle \cdot \langle 1, -3 \sin t, 3 \cos t \rangle \, dt \\ &= \int_0^\pi 3 \sin t - 9 \cos t \sin t - 3t \cos t \, dt \\ \int_0^\pi 3 \sin t \, dt &= [-3 \cos t]_{t=0}^{t=\pi} = 6 \\ \int_0^\pi 9 \cos t \sin t \, dt &= \left[\frac{9 \sin^2 t}{2} \right]_{t=0}^{t=\pi} = 0 \\ \int_0^\pi 3t \cos t \, dt &= [3t \sin t]_{t=0}^{t=\pi} - \int_0^\pi 3 \sin t \, dt \\ &= 0 + [3 \cos t]_{t=0}^{t=\pi} \\ &= -6 \\ \Rightarrow \int_C \mathbf{F} \cdot d\mathbf{r} &= 6 - 0 - (-6) = 12 \end{aligned}$$